

Part A: Probability & Conceptual Foundation

1. What is Conditional Probability?

Conditional probability is the probability of an event occurring given that another event has already happened.

It is denoted as **$P(A|B)$** , meaning probability of A given B.

Formula: **$P(A|B) = P(A \cap B) / P(B)$** .

Used when events are dependent on each other.

2. Explain Bayes' Theorem and its importance in classification problems.

Bayes' Theorem calculates the probability of a class given some features.

Formula: **$P(A|B) = [P(B|A) \times P(A)] / P(B)$** .

In classification, it helps predict the class label based on prior and likelihood.

Used in algorithms like **Naive Bayes** for probabilistic classification.

3. What assumptions does the Naive Bayes Classifier make?

Naive Bayes assumes that all features are **independent** given the class label.

This means one feature does not affect another.

It also assumes equal importance of features.

Though unrealistic, it works well in many real-world cases.

4. Explain the working principle of:

• K-Nearest Neighbors (KNN):

KNN classifies a data point based on the majority class of its **K nearest neighbors**.

Distance metrics like Euclidean distance are used.

It is a **lazy learning algorithm** (no training phase).

• Support Vector Machine (SVM):

SVM finds the **optimal hyperplane** that separates classes with maximum margin.

It focuses on boundary points called **support vectors**.

Can handle non-linear data using kernels.

5. Compare distance-based, probabilistic, and margin-based classifiers.

- **Distance-based (KNN):**
Uses distance to classify data points; simple but slow for large data.
- **Probabilistic (Naive Bayes):**
Uses probability and assumptions; fast and works well with text data.
- **Margin-based (SVM):**
Uses decision boundary with maximum margin; powerful but computationally expensive.